Mohamed Rayan Barhdadi

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Research Interests

1. Physics-informed and cognitive-science approaches to vision
2. 3D/4D Scene understanding and reconstruction
3. Vision-Language Models (VLMs) and video understanding

Education

Texas A&M University 2023 – 2027
Bachelor of Science, Electrical Engineering – *(Qatar Campus)*

- *Thesis (3rd year):* "The Physical World as the First Teacher: Structured 4D Scene Representation." *Advisors:* Prof. Hasan Kurban *and* Prof. Erchin Serpedin.
- *Notable Award*:
  - Richard E. Ewing Award for Excellence in Student Research (1 out of all students; awarded to one undergraduate annually by faculty nomination).
- *Selected Summer/Winter Schools* (typically for PhD students):
  - International Computer Vision Summer School (ICVSS), Italy, 2026.
  - African Computer Vision Summer School (ACVSS), Ghana, 2026.
  - London Geometry and Machine Learning (LOGML), England, 2026.

Publications (Google Scholar)

** denotes equal contribution*

IRIS: A Real-World Benchmark for Inverse Recovery and Identification of Physical Dynamic Systems from Monocular Video
R Khanbayov*, MR Barhdadi*, E Serpedin, and H Kurban
European Conference on Computer Vision ECCV, 2026

EMPATHIA: Multi-Faceted Human-AI Collaboration for Refugee Integration
MR Barhdadi, M Tuncel, E Serpedin, and H Kurban
Neural Information Processing Systems (Main Proceedings, CreativeAI) NeurIPS, 2025
+ Invited Keynote Talk at NeurIPS Affinity Workshop 2025

PhysicsNeRF: Physics-Guided 3D Reconstruction from Sparse Views
MR Barhdadi, H Kurban, and H Alnuweiri
Building Physically Plausible World Models Workshop
International Conference on Machine Learning ICML–W, 2025

Under Review

4D Synchronized Fields: Motion-Language Gaussian Splatting
MR Barhdadi, S Abdaljalil, R Khanbayov, E Serpedin, and H Kurban
AAAI Conference on Artificial Intelligence AAAI, 2027 (under rev.)

Intrinsic 4D Gaussian Segmentation from Scene Cues
H Yazar*, MR Barhdadi*, E Serpedin, M Tuncel, and H Kurban
British Machine Vision Conference BMVC, 2026 (under rev.)

Perception, Not Reasoning, Limits Visual Theory of Mind
MR Barhdadi, ST Wasim, J Gall, E Serpedin, and H Kurban
Neural Information Processing Systems NeurIPS, 2026 (under rev.)
Also Initial Version Accepted at 2nd Workshop on MMRAgI CVPR–W, 2026

Mixture Normalizing Flows with Exact Likelihood
MR Barhdadi, SB Belhaouari, H Bensmail, and M Aupetit
Neural Information Processing Systems NeurIPS, 2026 (under rev.)

Research Experience

Harvard University, AI and Robotics Lab 03/2026 – Present
Research Assistant with Prof. Mengyu Wang and Dr. Minghan Li
Topic: 3D/4D Understanding, Vision Language Models.

Texas A&M Qatar, Kurban Intelligence Lab 02/2025 – Present
Research Assistant with Prof. Hasan Kurban and Prof. Erchin Serpedin
Topics: All things Perception/Vision from Cog Sci/Phys approach.

Qatar Computing Research Institute, AI Center 05/2025 – 03/2026
Research Intern with Dr. Halima Bensmail and Dr. Michael Aupetit
Topics: Normalizing Flows and Generative Models.

Texas A&M Qatar, Power System Modeling Laboratory 01/2024 – 12/2024
Research Assistant with Prof. Selma Awadallah
Topic: ML for fault detection in electrical transformers.

Industry Experience

Overshoot (Y Combinator W26) 05/2026 – Present
Research Scientist Intern with Younes El Hjouji and Zakaria El Hjouji
Topic: Unified Encoder for Vision Language Models.

Ottonomi AI 07/2025 – 10/2025
Computer Vision Software Engineer with Prof. Hussein Alnuweiri
Under NDA ;)

Digital & Integration, SLB 06/2025 – 08/2025
Engineering Intern with Azhar Khan
Topic: Automated Well Integrity Evaluation.

Grants and Funding

ValuesLab Award Grant, Columbia University 2026 – 2027
Principal Investigator, Awarded $3,000 of funding and compute.
Topic: Machine Imagination as a new learning paradigm.

Student Research Experience Grants (SREG), TAMUQ 2026 – 2027
Lead Undergraduate Investigator, Awarded $50,000 of funding.
Topic: Physics informed 3D reconstruction and understanding.

Undergraduate Research Experience Project (UREP), QRDI 2025 – 2026
Member, Awarded $26,000 of funding.
Topic: ML for fault detection in electrical transformers.

Travel Award to ICVSS in Sicily, Italy, TAMUQ 2026
Travel Award to CVPR in Denver, USA, Overshoot (YC W26) and TAMUQ 2026
Travel Award to ACVSS in Accra, Ghana, Google and INRIA 2026
Travel Award to EACL in Rabat, Morocco, TAMUQ 2026
Travel Award to MenaML in Jeddah, Saudi, KAUST and Google DeepMind 2026
Travel Award to NeurIPS in San Diego, USA, TAMUQ 2025
Travel Award to ICML in Vancouver, Canada, TAMUQ 2025
Travel Funding to present at IFTP in College Station, USA, TAMU 2025
Travel Award for SLEP leadership exchange in College Station, USA, TAMU 2025

Complementary Registration to LOGML in London, England, Imperial College 2026
Complementary Registration NeurIPS, NeurIPS Committee 2025

Selected Talks

Invited Keynote Speaker at the "MusIML" Workshop, NeurIPS 2025 12/2025
Invited Speaker at "Human Inspired Computer Vision" Seminar, TAMUQ 11/2025
Selected Speaker at the "Undergraduate Research Conference", MIT 10/2025
Invited Speaker at Moroccans in Artificial Intelligence Research (MAIR) 10/2025

Honors and Awards

Richard E. Ewing Award for Excellence in Research (1 out of ~450) – $1,250 2026
Undergraduate Research Scholars, College of Engineering, TAMU 2026
1st Place at Qatar Computing Research Institute Symposium (1/108) 2025
First Place at AIX Hackathon, Rising Ventures (+85 Teams) – $5,000 2025
1st Place at EC 3rd Undergraduate Research Retreat – $550 2025
Woqod x Qatar Foundation Scholar Award Recipient – $10,000 2025
Second Place (Global Phase) Invent for the Planet by TAMU – $2,500 2025
First Place (Qatar Phase) Invent for the Planet by TAMUQ – $1,650 2024
+Best Prototype Award
+Best Video Award
Winner of Qatar Foundation Tech Pitch Competition – $11,000 2024
2nd Place Texas A&M University Qatar Robotics Competition 2024
Inducted in Engineering Honors Program at TAMUQ 2023

Academic Service

Reviewer, NeurIPS, CreativeAI Track 2026

Reviewer, ICML, AI4Good Workshop 2026
Reviewer, CVPR, ReLearn: Rediscovering Intelligence Workshop 2026
Volunteer, ICML/NeurIPS, MusIML Workshop 2025

Leadership

ACL, *Student Member* 2026 – Present
IEEE, *Student Member* 2023 – Present
The Peace Club TAMUQ, *Advisor* 2025
The Peace Club TAMUQ, *President* 2025
The Peace Club TAMUQ, *Vice-President* 2024
Lead Organizer, "Effective Humanitarian Engineering Solutions Workshop" 2024
Qatar Foundation, *Student Orientation Leader* 2024
IEEE Student Chapter, *Class Representative* 2023
Moroccan National Swimming Federation, *Instructor Volunteer* 2022

Languages

English (Fluent) · Arabic (Native) · French (Bilingual Proficiency)
Spanish (Elementary)

References

Hasan Kurban hkurban@hbku.edu.qa
Assistant Professor, Hamad Bin Khalifa University
Adjunct Associate Professor, Indiana University Bloomington

Erchin Serpedin eserpedin@tamu.edu
Department Chair, ECEN, Texas A&M University Qatar
Full Professor, Texas A&M University

Halima Bensmail hbensmail@hbku.edu.qa
Principal Scientist, Qatar Computing Research Institute
Previously: Oak Ridge National Lab, University of Washington, INRIA

Zakaria El Hajouji zakaria@overshoot.ai
Founder & CEO, Overshoot (Y Combinator W26)
Previously: MIT, Meta, Uber